Problem Set 3

Emily Zhang

Due: November 12, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 12, 2026. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```
model1 <- lm(voteshare ~ difflog, data = inc.sub)
summary(model1)
output_stargazer("q1_regression.tex", model1,
                 title = "Regression of voteshare on difflog",
                 label = "tab:q1")
```

Table 1: Regression of voteshare on difflog

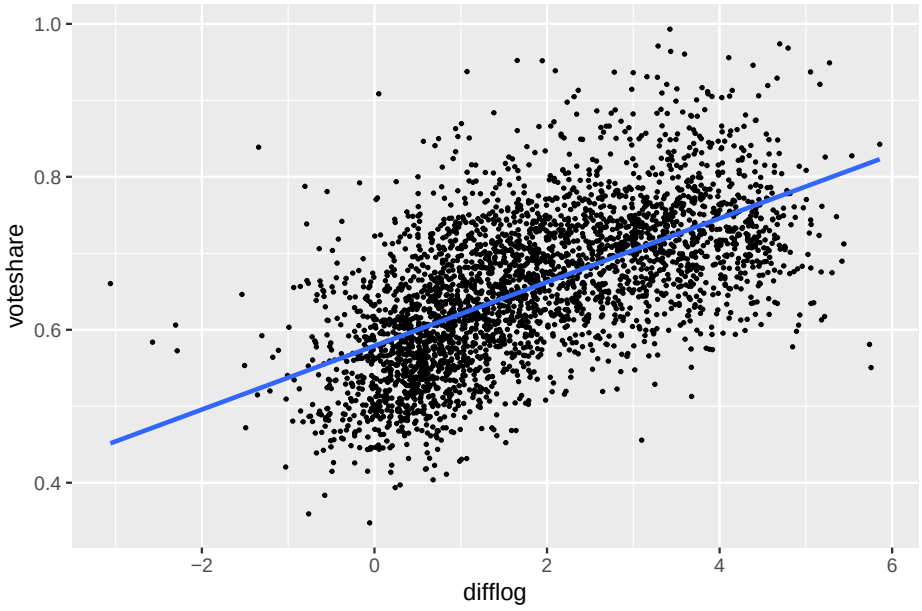
	Dependent variable:
	voteshare
difflog	0.042***
	(0.001)
Constant	0.579***
	(0.002)
Observations	3,193
R^2	0.367
Adjusted R^2	0.367
Residual Std. Error	0.079 (df = 3191)
F Statistic	1,852.791*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

Analysis:

2. Make a scatterplot of the two variables and add the regression line.

```
plot1 <- ggplot(inc.sub, aes(x=difflog, y=voteshare)) +
  geom_point(size = 0.5) +
  geom_smooth(method = "lm", se = FALSE)
ggsave("q1_plot.pdf", plot1, width = 6, height = 4)
```

Figure 1: Incumbent vote share vs. difference in campaign spending (log).



Analysis:

3. Save the residuals of the model in a separate object.

```
residuals1 <- residuals(model1)
summary(residuals1)
```

```
    Min.    1st Qu.     Median       Mean    3rd Qu.       Max.
-0.268319 -0.053454 -0.003769  0.000000  0.047798  0.327488
```

Analysis:

4. Write the prediction equation.

```
intercept1 <- round(coef(model1)[1], 3)
slope1 <- round(coef(model1)[2], 3)
equation1 <- paste("voteshare =", intercept1, "+", slope1, "* difflog")
equation1
```

$$\widehat{\text{voteshare}} = 0.579 + 0.042 \times \text{difflog}$$

Analysis:

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

```
model2 <- lm(presvote ~ difflog, data = inc.sub)
summary(model2)
output_stargazer("q2_regression.tex", model2,
                 title = "Regression of presvote on difflog",
                 label = "tab:q2")
```

Table 2: Regression of presvote on difflog

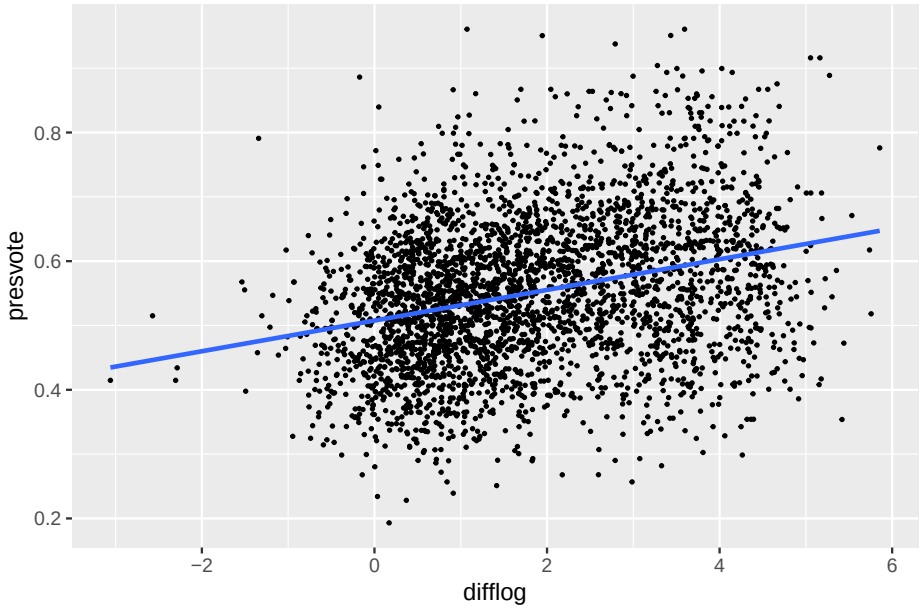
	Dependent variable:
	presvote
difflog	0.024***
	(0.001)
Constant	0.508***
	(0.003)
Observations	3,193
R^2	0.088
Adjusted R^2	0.088
Residual Std. Error	0.110 (df = 3191)
F Statistic	307.715*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

Analysis:

2. Make a scatterplot of the two variables and add the regression line.

```
plot2 <- ggplot(inc.sub, aes(x=difflog, y=presvote)) +
  geom_point(size = 0.5) +
  geom_smooth(method = "lm", se = FALSE)
ggsave("q2_plot.pdf", plot2, width = 6, height = 4)
```

Figure 2: Presidential vote share vs. difference in campaign spending (log).



Analysis:

3. Save the residuals of the model in a separate object.

```
residuals2 <- residuals(model2)
summary(residuals2)
```

```
    Min.  1st Qu.   Median     Mean  3rd Qu.     Max.
-0.321965 -0.074069 -0.001018  0.000000  0.071507  0.427435
```

Analysis:

4. Write the prediction equation.

```
intercept2 <- round(coef(model2)[1], 3)
slope2 <- round(coef(model2)[2], 3)
equation2 <- paste("presvote =", intercept2, "+", slope2, "* difflog")
equation2
```

$$\widehat{\text{presvote}} = 0.508 + 0.024 \times \text{difflog}$$

Analysis:

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```
model3 <- lm(voteshare ~ presvote, data = inc.sub)
summary(model3)
output_stargazer("q3_regression.tex", model3,
                 title = "Regression of voteshare on presvote",
                 label = "tab:q3")
```

Table 3: Regression of voteshare on presvote

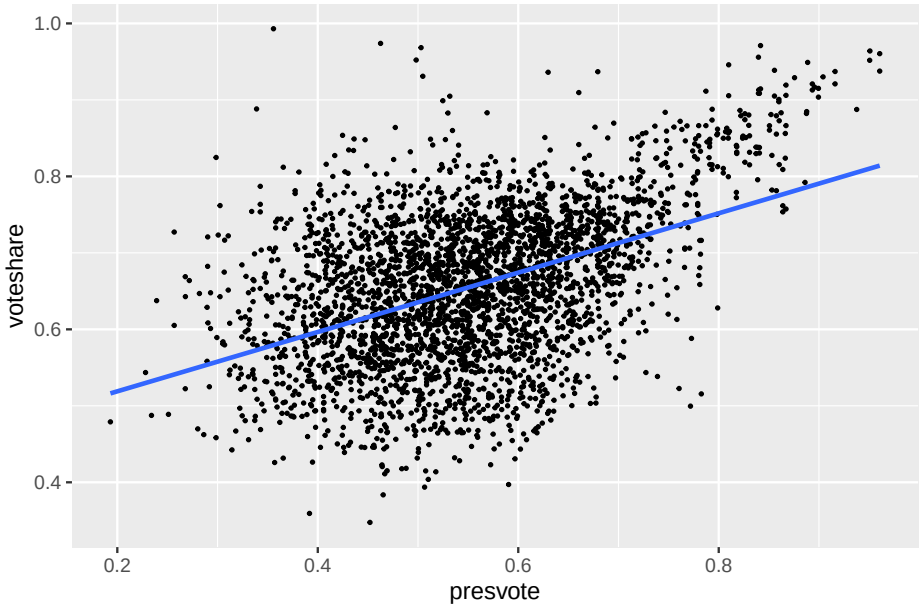
	Dependent variable:
	voteshare
presvote	0.388***
	(0.013)
Constant	0.441***
	(0.008)
Observations	3,193
R^2	0.206
Adjusted R^2	0.206
Residual Std. Error	0.088 (df = 3191)
F Statistic	826.950*** (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

Analysis:

2. Make a scatterplot of the two variables and add the regression line.

```
plot3 <- ggplot(inc.sub, aes(x=presvote, y=voteshare)) +
  geom_point(size = 0.5) +
  geom_smooth(method = "lm", se = FALSE)
ggsave("q3_plot.pdf", plot3, width = 6, height = 4)
```

Figure 3: Incumbent vote share vs. presidential vote share.



Analysis:

3. Write the prediction equation.

```
intercept3 <- round(coef(model3)[1], 3)
slope3 <- round(coef(model3)[2], 3)
equation3 <- paste("voteshare =", intercept3, "+", slope3, "* presvote")
equation3
```

$$\widehat{\text{voteshare}} = 0.441 + 0.388 \times \text{presvote}$$

Analysis:

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
residuals <- data.frame(residuals1, residuals2)
model4 <- lm(residuals1 ~ residuals2, data = residuals)
summary(model4)
output_stargazer("q4_regression.tex", model4,
                 title = "Regression of Q1 residuals on Q2 residuals",
                 label = "tab:q4")
```

Table 4: Regression of Q1 residuals on Q2 residuals

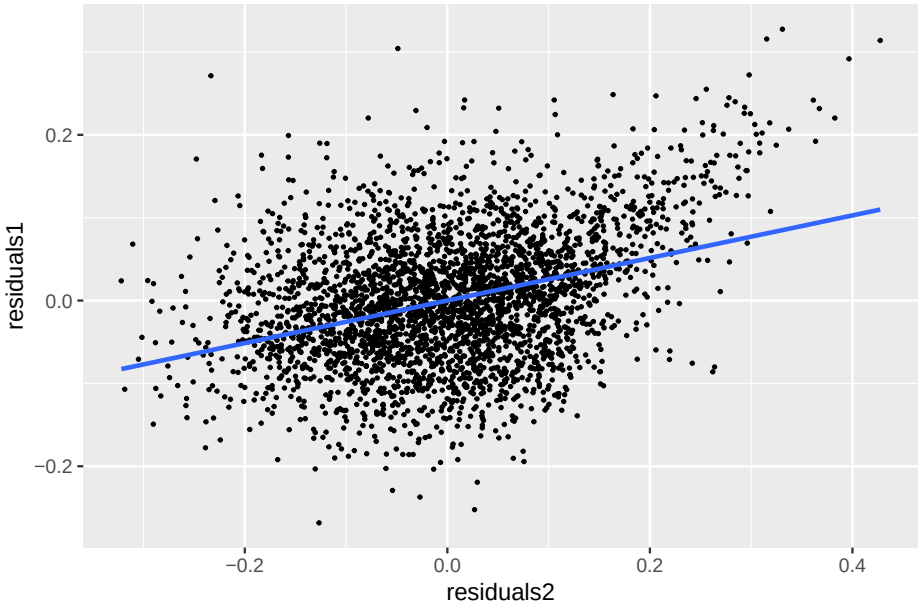
	Dependent variable:
	residuals1
residuals2	0.257^{***}
	(0.012)
Constant	-0.000
	(0.001)
Observations	3,193
R^2	0.130
Adjusted R^2	0.130
Residual Std. Error	0.073 (df = 3191)
F Statistic	476.975^{***} (df = 1; 3191)
Note:	*p<0.1; **p<0.05; ***p<0.01

Analysis:

2. Make a scatterplot of the two residuals and add the regression line.

```
plot4 <- ggplot(residuals, aes(x=residuals2, y=residuals1)) +
  geom_point(size = 0.5) +
  geom_smooth(method = "lm", se = FALSE)
ggsave("q4_plot.pdf", plot4, width = 6, height = 4)
```

Figure 4: Added-variable plot: residuals of `voteshare` vs. residuals of `presvote` (both on `difflog`).



Analysis:

3. Write the prediction equation.

```
intercept4 <- round(coef(model4)[1], 3)
slope4 <- round(coef(model4)[2], 3)
equation4 <- paste("voteshare_residual =", intercept4, "+", slope4, "*
    presvote_residual")
equation4
```

$$\widehat{\text{voteshare residual}} = 0 + 0.257 \times \text{presvote residual}$$

Analysis:

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
model5 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
summary(model5)
output_stargazer("q5_regression.tex", model5,
                 title = "Regression of voteshare on difflog and presvote
    ",
                 label = "tab:q5")
```

Table 5: Regression of voteshare on difflog and presvote

	Dependent variable:
	voteshare
difflog	0.036***
	(0.001)
presvote	0.257***
	(0.012)
Constant	0.449***
	(0.006)
Observations	3,193
R^2	0.450
Adjusted R^2	0.449
Residual Std. Error	0.073 (df = 3190)
F Statistic	1,302.947*** (df = 2; 3190)
Note:	*p<0.1; **p<0.05; ***p<0.01

Analysis:

2. Write the prediction equation.

```
intercept5 <- round(coef(model5)[1], 3)
slope5_difflog <- round(coef(model5)[2], 3)
slope5_presvote <- round(coef(model5)[3], 3)
```

```
equation5 <- paste("voteshare =", intercept5, "+", slope5_difflog, "*
    difflog", "+", slope5_presvote, "* presvote")
equation5
```

$$\widehat{\text{voteshare}} = 0.449 + 0.036 \times \text{difflog} + 0.257 \times \text{presvote}$$

Analysis:

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

   Analysis: